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Image processing system, image processing method, and storage medium

Abstract

An image processing system includes an image acquisition unit configured to acquire an image signal generated by an imaging device that captures an optical image having a low-distortion region and a high-distortion region, a setting unit configured to set a distortion-correction region on which distortion-correction is performed for the image signal and a non-distortion-correction region on which distortion-correction is not performed for the image signal on the basis of characteristics of the optical image; and a display signal generation unit configured to perform distortion-correction for the image signal of the distortion-correction region on the basis of the characteristics of the optical image, and generate a synthesized image by synthesizing the image signal on which distortion-correction has been performed and the image signal of the non-distortion-correction region.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to an image processing system capable of correcting distortion, an image processing method, and a storage medium.

Description of the Related Art

(2) In recent years, there has been a demand to replace a rearview mirror mounted on a vehicle with an electronic rearview mirror. For example, Japanese Patent Laid-Open No. 2010-95202 discloses an electronic rearview mirror system constituted by an imaging unit configured to have an imaging range to the rear of and outside a vehicle and a display unit in the vehicle, and configured to allow a driver to check a state to the rear of and outside the vehicle by displaying an image captured by the imaging unit on a display inside the vehicle.

(3) On the other hand, there is a rearward checking system that allows a driver to check a blind area behind a vehicle when the vehicle moves backward. Japanese Patent Laid-Open No. 2004-345554 discloses a rearward checking system for allowing a driver to check a blind area behind a vehicle at the time of moving backward by installing a camera to image to the rear of the vehicle and displaying the captured image in an interior of the vehicle.

(4) A camera as the imaging unit for capturing an image for the above-described electronic rearview mirror is required to have a high-resolution for allowing a driver to check a state of a relatively distant place to the rear more finely. On the other hand, a rearward checking camera system is required to capture a wider range to confirm safety in a wider range including a blind area behind the vehicle and a rear lateral side to avoid a collision at the time of moving backward or the like.

(5) Also, when an electronic rearview mirror system and a rearward checking system are mounted on a vehicle at the same time, if a camera for the electronic rearview mirror system and a camera for the rearward checking system are separately mounted, an in-vehicle image processing system becomes complicated. Such a problem also occurs in, for example, an automatic driving system in which a plurality of cameras is disposed to image a state around the vehicle to perform automatic driving or the like.

(6) On the other hand, the number of cameras installed in a vehicle can be reduced by employing, for example, a camera that uses a special ultra-wide-angle lens. However, if an ultra-wide-angle lens or the like is used, although a wide angle of view can be obtained, distortion in the peripheral portion is large, and it is difficult to recognize an object in the peripheral portion unless the distortion is corrected. On the other hand, a delay time is incurred if the screen is displayed after the distortion is corrected, causing a delay in image recognition or visual recognition of an obstacle or the like, and this may cause a danger.

(7) In view of the above-described problems, one objective of the present invention is to provide an image processing system suitable for early recognition of an obstacle or the like.

SUMMARY OF THE INVENTION

(8) In order to achieve the above-described objective, an image processing system relating to one aspect of the present invention includes at least one processor or circuit configured to function as: an image acquisition unit configured to acquire an image signal generated by an imaging device that captures an optical image having a low-distortion region and a high-distortion region; a setting unit configured to set a distortion-correction region on which distortion-correction is performed for the image signal and a non-distortion-correction region on which distortion-correction is not performed for the image signal on the basis of characteristics of the optical image; and a display signal generation unit configured to perform distortion-correction for the image signal of the distortion-correction region on the basis of the characteristics of the optical image, and generate a synthesized image by synthesizing the image signal on which distortion-correction has been performed and the image signal of the non-distortion-correction region.

(9) Further features of the present invention will become apparent from the following description of embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a view illustrating a positional relationship between a vehicle and an imaging unit in an image processing system of a first embodiment.

(2) FIGS. 2A and 2B are diagrams for explaining optical characteristics of the imaging unit of the first embodiment.

(3) FIG. 3 is a functional block diagram for explaining a configuration of the image processing system of the first embodiment.

(4) FIG. 4 is a diagram for explaining a correspondence relationship between an angle of view, a high-resolution region, and a low-resolution region of a camera unit **13** on a rear side in the first embodiment.

(5) FIG. 5A is a diagram showing an example of an image signal obtained from an imaging element when distortion-correction is not performed, and FIG. 5B is a diagram for explaining an image signal after distortion-correction.

(6) FIG. 6A is a diagram showing an example in which distortion is corrected for an entire screen, FIG. 6B is a diagram showing an example of synthesizing images by performing distortion-correction on an image of the low-resolution region without performing distortion-correction on an image of the high-resolution region, and FIG. 6C is a diagram showing an example of a case in which distortion-correction is not performed for the entire screen.

(7) FIG. 7 is a flowchart for explaining an example of a case in which a pattern of FIG. 6 is manually switched.

(8) FIG. 8 is a flowchart for explaining a distortion region selection example in a second embodiment.

(9) FIG. 9 is a flowchart for explaining a distortion region selection example in a third embodiment.

(10) FIGS. 10A to 10E are diagrams for explaining examples of a distortion-correction region, a display region, and an image recognition region in the third embodiment.

(11) FIGS. 11A to 11D are diagrams for explaining a display example of a first display unit in the third embodiment.

(12) FIG. 12A is a diagram for explaining a display example of a display screen **501** of a first display unit **50** of the third embodiment, and FIG. 12B is a diagram for explaining a display example of a display screen **511** of a second display unit **51** of the third embodiment.

(13) FIG. 13A is a diagram for explaining a display example of an image **85B** having a narrow

angle of view in a vertical direction at the time of moving backward in the third embodiment, and FIG. 13B is a diagram showing a display example of an image 86B having a wide angle of view for backward movement in the third embodiment.

DESCRIPTION OF THE EMBODIMENTS

(14) Hereinafter, with reference to the accompanying drawings, favorable modes of the present invention will be described using Embodiments. In each diagram, the same reference signs are applied to the same members or elements, and duplicate description will be omitted or simplified.

First Embodiment

(15) In a first embodiment, an improved method in which a high-definition display for an electronic rearview mirror and a display for checking surroundings of a vehicle such as a wide range of the rear are achieved at the same time with a small number of cameras will be described. FIG. 1 is a view illustrating a positional relationship between a vehicle and imaging units in an image processing system of the first embodiment.

(16) In the first embodiment, as illustrated in FIG. 1, camera units 11, 12, 13, and 14 are installed on a front side, a right side, a rear side, and a left side of, for example, a vehicle 1 which is an automobile as a mobile object. Further, four camera units are used in the first embodiment, but the number of camera units is not limited to four, and may be at least one or more camera units.

(17) The camera units 11 to 14 are installed to have an imaging range covering in front of, a right side of, a left side of, and to the rear of the vehicle 1 as the mobile object.

(18) The camera units 11 to 14 have substantially the same configuration as each other, and each have an imaging element capturing an optical image, and an optical system configured to form an optical image on a light receiving surface of the imaging element. Further, for example, optical axes of the optical systems of the camera units 11 and 13 are installed to be substantially horizontal, and optical axes of the optical systems of the camera units 12 and 14 are installed to face slightly downward from the horizontal.

(19) Also, the optical systems of the camera units 11 to 14 used in the first embodiment are each configured so that a high-definition image can be obtained at a narrow angle of view around the optical axis, and a low-resolution captured image can be obtained in a wide angle of view. Further, reference signs 11a to 14a denote imaging angles of view in which a high-resolution and low-distortion image can be captured, and reference signs 11b to 14b denote imaging angles of view in which a low-resolution and high-distortion image can be captured.

(20) The optical systems of the camera units 11 to 14 in the first embodiment will be described with reference to FIG. 2. Further, characteristics of the optical systems of the camera units 11 to 14 do not have to be the same, but in the first embodiment, it is assumed that the optical systems of the camera units 11 to 14 have substantially the same characteristics, and an optical system of the camera unit 11 will be described as an example.

(21) FIGS. 2A and 2B are diagrams for explaining optical characteristics of the imaging unit in the first embodiment of the present invention, and FIG. 2A is a diagram in which an imaged height y at each half angle of view on a light receiving surface of the imaging element in the optical system of the camera unit 11 in the first embodiment is shown in a contour line.

(22) FIG. 2B is a diagram showing a projection characteristic representing a relationship between the imaged height y and a half angle of view θ of the optical system of the camera unit 11 in the first embodiment. In FIG. 2B, the half angle of view (angle formed by an optical axis and incident light) θ is taken as a horizontal axis, and an image forming height (imaged height) y on a sensor surface (imaging plane) of the camera unit 11 is taken as a vertical axis.

(23) As shown in FIG. 2B, the optical system of the camera unit 11 in the first embodiment is configured so that a projection characteristic $y(\theta)$ differs between a region having a half angle of view less than a predetermined half angle of view θ_a and a region having a half angle of view equal to or greater than the half angle of view θ_a . Therefore, when an amount of increase in the imaged height y with respect to the half angle of view θ per unit is referred to as a resolution, the resolution

differs depending on regions.

(24) It can be said that this local resolution is expressed by a differential value $dy(\theta)/d\theta$ of the projection characteristic $y(\theta)$ with respect to the half angle of view θ . That is, it can be said that the resolution becomes higher as an inclination of the projection characteristic $y(\theta)$ in FIG. 2B becomes larger. Also, it can also be said that the resolution becomes higher as a distance between the imaged heights y at the contour line-shaped half angle of view in FIG. 2A becomes larger.

(25) In the first embodiment, a region close to a center formed on the sensor surface when the half angle of view θ is less than the predetermined half angle of view θ_a is called a high-resolution region **10a**, and an outer region when the half angle of view θ is equal to or greater than the predetermined half angle of view θ_a is called a low-resolution region **10b**. Further, the high-resolution region **10a** corresponds to the imaging angle of view **11a**, and the low-resolution region **10b** corresponds to the imaging angle of view **11b**.

(26) Further, in the first embodiment, a circle at a boundary between the high-resolution region **10a** and the low-resolution region **10b** is called a resolution boundary, and a boundary image on a display screen corresponding to the resolution boundary is called a display resolution boundary or simply a boundary image. Further, the boundary image (display resolution boundary) displayed on the display screen may not be circular. It may be rectangular or the like for convenience.

(27) Further, in the first embodiment, the high-resolution region **10a** is a low-distortion region with a relatively small distortion, and the low-resolution region **10b** is a high-distortion region with a relatively large distortion. Therefore, in the first embodiment, the high-resolution region and the low-resolution region correspond to the low-distortion region and the high-distortion region, and the high-resolution region and the low-resolution region may be referred to as the low-distortion region and the high-distortion region. Also, conversely, the low-distortion region and the high-distortion region may be referred to as the high-resolution region and the low-resolution region.

(28) The optical system of the camera unit **11** in the first embodiment is configured so that the projection characteristic $y(\theta)$ thereof is larger than $f \times \theta$ in the high-resolution region (low-distortion region) **10a** (f is afocal length of the optical system of the camera unit **11**). Also, the projection characteristic $y(\theta)$ in the high-resolution region (low-distortion region) is set to be different from the projection characteristic in the low-resolution region (high-distortion region).

(29) Also, when $\theta_{\max}$ is a maximum half angle of view of the optical system of the camera unit **11**, a ratio $\theta_a/\theta_{\max}$ of θ_a and $\theta_{\max}$ is preferably a predetermined lower limit value or higher and, for example, 0.15 to 0.16 is preferable as the predetermined lower limit value.

(30) Also, the ratio $\theta_a/\theta_{\max}$ of θ_a to $\theta_{\max}$ is preferably a predetermined upper limit value or lower and is preferably, for example, 0.25 to 0.35. For example, if $\theta_{\max}$ is 90° , the predetermined lower limit value is 0.15, and the predetermined upper limit value is 0.35, it is desirable to determine **6a** in a range of 13.5 to 31.5° .

(31) Further, the optical system of the camera unit **11** is configured so that the projection characteristic $y(\theta)$ thereof satisfies the following equation 1.

(32)
$$1 < \frac{f \times \sin \theta_{\max}}{y(\theta_{\max})} \leq A \quad (\text{equation 1})$$

(33) f is the focal length of the optical system of the camera unit **11** as described above, and A is a predetermined constant. When the lower limit value is set to 1, a center resolution can be made higher than that of a fisheye lens of an orthographic projection method ($y=f \times \sin \theta$) having the same maximum imaged height, and when the upper limit value is set to A , satisfactory optical performance can be maintained while obtaining an angle of view equivalent to that of the fisheye lens. The predetermined constant A may be determined in consideration of a balance between resolutions of the high-resolution region and the low-resolution region, and is preferably 1.4 to 1.9.

(34) With the optical system configured as described above, an amount of increase in the imaged height y with respect to the half angle of view θ per unit is reduced in the low-resolution region **10b** while a high-resolution can be obtained in the high-resolution region **10a**, and thus a wider angle of

view can be captured. Therefore, a high-resolution can be obtained in the high-resolution region **10a** while having a wide angle of view equivalent to that of the fisheye lens as the imaging range. (35) Further, in the first embodiment, since the high-resolution region (low-distortion region) has a characteristic similar to that of a central projection method ($y=f\times\tan\theta$) or an equidistant projection method ($y=f\times\theta$) having a projection characteristic of a normal optical system for imaging, an optical distortion is small and display can be made finely in the high-resolution region (low-distortion region). Therefore, a natural perspective can be obtained when visually observing surrounding vehicles or the like such as a preceding vehicle or a following vehicle, and satisfactory visibility can be obtained by suppressing deterioration of image quality.

(36) Further, since the same effects can be obtained as long as the projection characteristic $y(\theta)$ satisfies the above-described equation (1), the present invention is not limited to the projection characteristic shown in FIG. 2. Further, in the first embodiment, the optical system having the projection characteristic $y(\theta)$ satisfying the above-described equation (1) may be referred to as a different-angle-of-view lens.

(37) Further, the high-resolution regions **10a** of the optical systems of the camera units **11** to **14** correspond to the imaging angles of view **11a** to **14a**, and the low-resolution regions **10b** thereof correspond to the imaging angles of view **11b** to **14b**.

(38) Next, a configuration of the image processing system in the first embodiment will be described with reference to FIG. 3. FIG. 3 is a functional block diagram for explaining a configuration of the image processing system of the first embodiment.

(39) In FIG. 3, an image processing system **100** is mounted on the vehicle **1** as the mobile object, and imaging units **21** to **24** and camera processing units **31** to **34** are disposed in housings of the camera units **11** to **14**.

(40) The imaging units **21** to **24** include different-angle-of-view lenses **21c** to **24c** and imaging elements **21d** to **24d** such as CMOS image sensors or CCD image sensors. Here, the imaging units **21** to **24** function as image acquisition units, and the image acquisition units each acquires an image signal generated by the imaging unit that captures an optical image having a low-distortion region and a high-distortion region. That is, an image acquisition step is included therein. Also, a plurality of image acquisition units disposed at different positions are used in the first embodiment.

(41) The different-angle-of-view lenses **21c** to **24c** as the optical system are each configured by one or more optical lenses, have the projection characteristic $y(\theta)$ satisfying the condition of equation (1), and form optical images having a low-distortion region and a high-distortion region on the light receiving surfaces of the imaging elements **21d** to **24d**. The imaging elements **21d** to **24d** each photoelectrically convert the optical image to output a captured image signal. For example, RGB color filters are disposed for each pixel on the light receiving surfaces of the imaging elements **21d** to **24d**. An RGB disposition is, for example, a Bayer disposition.

(42) Therefore, the imaging element is configured to sequentially output, for example, R, G, R, and G signals from a predetermined row, and G, B, G, and B signals from an adjacent row according to the Bayer disposition.

(43) Reference signs **31** to **34** denote the camera processing units, which are housed together with the imaging units **21** to **24** in the same housings of the camera units **11** to **14** and process the captured image signal output from the imaging units **21** to **24**. Further, in FIG. 3, details of the imaging unit **24**, the camera processing unit **34**, and wirings thereof are omitted for convenience.

(44) The camera processing units **31** to **34** include image processing units **31a** to **34a**, recognition units **31b** to **34b**, and camera information units **31c** to **34c**. The image processing units **31a** to **34a** process the captured image signal output from the imaging units **21** to **24**. Further, a part or all of the camera processing unit **31** may be performed by signal processing units laminated in the imaging elements **21d** to **24d**.

(45) Specifically, the image processing units **31a** to **34a** performs debayer processing on image data input from the imaging units **21** to **24** according to the Bayer disposition, and convert the

image data into RGB raster format image data. Further, correction processing of various types such as white balance adjustment, gain/offset adjustment, gamma processing, color matrix processing, and lossless compression processing are performed. However, a so-called RAW image signal is formed without performing lossy compression processing or the like.

(46) The recognition units **31b** to **34b** (first image recognition unit) each recognize an image of each predetermined object (for example, an automobile, a person, an obstacle, or the like) from non-distortion-corrected RAW image signals that have been image-processed by the image processing units **31a** to **34a**. That is, the recognition units **31b** to **34b** performs image recognition on the image signal corresponding to the low-distortion region in a state of the RAW image signal without distortion-correction, and output a first image recognition result.

(47) Further, the recognition units **31b** to **34b** of the first embodiment perform the image recognition processing for the RAW image signal obtained from at least the high-resolution region **10a**, and recognize the predetermined object. Therefore, the image recognition processing may be performed after cutting out only the high-resolution region **10a**.

(48) Further, at this time, the recognition units **31b** to **34b** may perform image recognition processing on the RAW image signal obtained from the low-resolution region **10b**. However, since the RAW image signal has not been corrected for distortion, an image in a peripheral portion of the different-angle-of-view lens has a large distortion and recognition reliability is lowered.

(49) Alternatively, the recognition units **31b** to **34b** may cut out the RAW image signal obtained from the high-resolution region **10a** and then perform the image recognition processing only on the cut-out RAW image signal obtained from the high-resolution region **10a**.

(50) Further, the region to be cut out for image recognition at that time is preferably a rectangle having a shape suitable for image recognition processing. Also, the rectangular region to be cut out may be only a part (for example, a rectangle inscribed in the high-resolution region **10a**) of the high-resolution region **10a**, or may be a rectangle including both the high-resolution region **10a** and the low-resolution region **10b**.

(51) Here, the recognition units **31b** to **34b** function as a first image recognition unit that performs image recognition on the basis of an image signal of at least a partial region among the image signals acquired by the imaging unit (image acquisition unit) to output the first image recognition result. Further, in the first embodiment, the partial region is a region corresponding to the low-distortion region.

(52) The recognition units **31b** to **34b** transmit types of the object and a set of coordinates to an integrated processing unit **40** as a recognition result.

(53) On the other hand, the recognition units **31b** to **34b** receive prediction information which is a set including types of the object and information on a moving direction of the object or information on a priority recognition region from an integrated control unit **41c** of the integrated processing unit **40**. This prediction information will be described later.

(54) Here, an output of the recognition unit **31b** of the camera unit **11** installed with respect to the front is directly supplied to a travel control unit (ECU) **60**. This is because it may be necessary to immediately stop traveling or control the traveling to avoid an obstacle on the basis of the recognition result of the recognition unit **31b** such as an obstacle.

(55) The camera information units **31c** to **34c** (characteristics information holding unit) hold camera information of the camera units **11** to **14** in a memory in advance. The camera information unit can also temporarily hold information from various sensors and the like provided in the camera units **11** to **14**. The camera information includes characteristic information (resolution boundary information or the like) of the optical image as shown in FIG. 2 formed by, for example, the different-angle-of-view lenses **21c** to **24c**.

(56) Also, the camera information includes the number of pixels of the imaging element **21d** to **24d**, information on mounting position coordinates and postures (pitch, roll, yaw, and the like) of the camera unit in vehicle coordinates, a direction of image capturing, and the like. The camera

information may include information such as gamma characteristics, sensitivity characteristics, and a frame rate.

(57) Further, the camera information may include information on an image processing method and an image format when the RAW image signal is generated in the image processing units **31a** to **34a**.

(58) Further, since a mounting position with respect to a vehicle is often determined for each camera unit, the mounting position coordinates may be stored in the memory in the camera information unit in advance. Also, posture coordinates of the camera unit are coordinates relative to the vehicle **1** and may be acquired from an encoder (not shown) or the like provided in the camera unit. Alternatively, the posture coordinates may be acquired using a three-dimensional acceleration sensor or the like.

(59) Also, information on the direction of image capturing may be acquired by using, for example, a geomagnetic sensor. Since the resolution boundary information of the camera is determined by a lens design, it is stored in the memory in the camera information unit in advance.

(60) Further, the camera information is information peculiar to the imaging units **21** to **24**, is different from each other, and those pieces of information are transmitted to the integrated processing unit **40** and is referred to when image processing or the like is performed in the integrated processing unit **40**. Here, the camera information units **31c** to **34c** function as a characteristics information holding unit that holds characteristic information on characteristics of an optical image.

(61) Further, a CPU as a computer and a memory for storing a computer program as a storage medium are incorporated in the camera processing units **31** to **34**. Also, the CPU is configured to control each unit in the camera processing units **31** to **34** by executing the computer program in the memory.

(62) Further, in the first embodiment, the image processing units **31a** to **34a** and the recognition units **31b** to **34b** use hardware such as a dedicated circuit (ASIC) or a processor (reconfigurable processor, DSP). Thereby, an increase in speed of image recognition in the high-resolution region can be realized, and a likelihood of avoiding an accident can be increased. Further, the image processing units **31a** to **34a** may have a distortion-correction function.

(63) Further, a part or all of functional blocks inside the camera processing units **31** to **34** may be realized by causing the CPU to execute a computer program stored in the memory, but in that case, it is desirable to increase a processing speed of the CPU.

(64) Reference sign **40** denotes the integrated processing unit, which includes a system on chip (SOC)/a field programmable gate array (FPGA) **41**, a CPU **42** as a computer, and a memory **43** as a storage medium. The CPU **42** performs various controls of the entire image processing system **100** by executing a computer program stored in the memory **43**. Further, in the first embodiment, the integrated processing unit **40** is housed in a housing separate from the camera unit.

(65) The SOC/FPGA **41** includes an image processing unit **41a**, a recognition unit **41b** (second image recognition unit), and the integrated control unit **41c**. The image processing unit **41a** acquires the RAW image signals from the camera processing units **31** to **34**, and acquires the camera information of the camera units **11** to **14** from the camera information units **31c** to **34c**.

(66) As described above, the camera information includes optical characteristics of the different-angle-of-view lenses **21c** to **24c**, the number of pixels, photoelectric conversion characteristics, gamma characteristics, and sensitivity characteristics of the imaging elements **21d** to **24d**, format information of the RAW image signal, information of mounting coordinates and postures of the camera unit in vehicle coordinates, and the like.

(67) The image processing unit **41a** performs resolution conversion for the RAW image signals from the camera processing units **31** to **34** on the basis of those pieces of camera information, and performs distortion-correction for the image signals obtained from the low-resolution regions **10b** of the imaging units **21** to **24**.

(68) That is, the image processing unit **41a** performs distortion-correction for the image signal of the distortion-correction region on the basis of characteristics of the optical image, and generates a synthesized image by synthesizing the image signal on which distortion-correction has been performed and the image signal of a non-distortion-correction region without distortion-correction. That is, the image processing unit **41a** also functions as a display signal generation unit, performs distortion-correction or the like, and performs a display signal generation step of generating a synthesized image. Further, in the first embodiment, the distortion-correction region can be set by a user or automatically.

(69) In the first embodiment, the image processing unit **41a** does not perform the distortion-correction because the image signal obtained from the high-resolution region **10a** has almost no distortion. However, the image processing unit **41a** may also perform simple distortion-correction on the image signal obtained from the high-resolution region **10a**. Also, the image processing unit **41a** performs lossy compression processing or the like on the RAW image signals sent from the camera processing units **31** to **34** as appropriate.

(70) Also, the image processing unit **41a** synthesizes the image signal of the low-resolution regions **10b** of the imaging units **21** to **24** to which distortion-correction has been performed and the image signal of the high-resolution region **10a** to be smoothly joined together to form the entire image for each of the imaging units **21** to **24**.

(71) Further, if distortion-correction is performed on both the image signal of the low-resolution region **10b** and the image signal obtained from the high-resolution region **10a**, the RAW image signals obtained by the image processing units **31a** to **34a** may be corrected for distortion as they are.

(72) The recognition unit **41b** performs image recognition processing with respect to the entire image for each of the imaging units **21** to **24** after performing distortion-correction on at least the low-resolution region, and recognizes an image of a predetermined object (for example, an automobile, a person, an obstacle, or the like) in the entire image for each of the imaging units **21** to **24**. That is, the recognition unit **41b** performs the image recognition after performing distortion-correction on the image signal corresponding to at least the low-resolution region (high-distortion region), and outputs a second image recognition result.

(73) At that time, recognition results (type and coordinates of the object) by the recognition units **31b** to **34b** are also referred to. Further, in the above description, the recognition unit **41b** performs image recognition with respect to the entire image for each of the imaging units **21** to **24**, but the image recognition for the entire image may not necessarily be performed. For example, it is not necessary to perform image recognition for a peripheral portion of the image.

(74) That is, the recognition unit **41b** may recognize, for example, a wider region including the regions recognized by the recognition units **31b** to **34b**.

(75) Here, the recognition unit **41b** functions as a second image recognition unit that performs image recognition on an image signal of a region wider than a partial region including the partial region on which image recognition has been performed by the first image recognition unit among the image signals acquired by the image acquisition unit to output the second image recognition result. Further, the second image recognition unit performs image recognition for a synthesized image obtained by synthesizing image signals corresponding to the high-resolution region **10a** as the low-distortion region and the low-resolution region **10b** as the high-distortion region to output the second image recognition result.

(76) In the first embodiment, the image processing unit **41a** synthesizes images from the camera units **12** to **14** as a plurality of imaging units to be smoothly joined together to form a panoramic synthesized image.

(77) In that case, it is desirable that images of the plurality of imaging units to be interfaced be set so that at least a part of each imaging angle of view has a predetermined amount or more of overlap region with each other.

(78) As will be described later, the camera units **12** and **13** are disposed so that imaging ranges overlap each other. Also, the camera units **13** and **14** are disposed so that imaging ranges overlap each other. Moreover, at that time, the camera units are disposed so that imaging ranges of the low-distortion regions of at least two image acquisition units overlap each other.

(79) Also, the recognition unit **41b** performs image recognition for a panoramic synthesized image thereof. Thereby, it is possible to recognize an image of an object imaged, for example, to straddle angles of view of the plurality of imaging units. This is because, the entire image of the object may not be known from the individual entire image from each imaging unit, but in the panoramic synthesized image, there are cases in which almost the entire of the object is reflected and the image recognition is possible by image processing.

(80) For example, if a recognition result by the recognition units **31b** to **34b** and a recognition result by the recognition unit **41b** are different, the integrated control unit **41c** outputs an integrated image recognition result by adopting a recognition result on a side with higher reliability.

(81) For example, a proportion of the object occupied in the image recognized by the recognition units **31b** to **34b** and a proportion of the same object occupied in a screen recognized by the recognition unit **41b** are compared, and a recognition result on a side having a larger proportion may be determined to have higher reliability and adopted.

(82) Alternatively, in a case of an object that straddles both the high-resolution region and the low-resolution region, the recognition result by the recognition unit **41b** may be determined to have higher reliability than the recognition result by the recognition units **31b** to **34b** and adopted. Alternatively, if a position of the object recognized by the recognition units **31b** to **34b** is at a peripheral portion of an image, the reliability may be determined to be low, and the recognition result by the recognition unit **41b** may be determined to have higher reliability and adopted.

(83) Alternatively, in the recognition unit **41b**, image recognition may be performed only on the low-resolution region with the low-resolution region corrected for distortion, and if there is an object that straddles the low-resolution region and the high-resolution region, image recognition may be performed for the object. That is, for an object that is present only in the high-resolution region, reliability of recognition by the recognition units **31b** to **34b** is considered to be high, and the recognition unit **41b** may be controlled not to perform the image recognition processing.

(84) Here, the integrated control unit **41c** functions as an integrated processing unit that outputs an image recognition result integrated on the basis of the reliability of the first image recognition result and the reliability of the second image recognition result.

(85) Also, the integrated control unit **41c** forms a signal for displaying a desired image among the entire image for each of the imaging units **21** to **24**, a panoramic synthesized image, and the like on a first display unit **50**, a second display unit **51**, and the like. Also, a frame for highlighting a recognized object, information on a type, a size, a position, a speed, or the like of the object, CG for warning, and the like are generated. Further, a CG of the boundary image for displaying the boundary is generated on the basis of characteristic information of the optical system such as display resolution boundary information acquired from the camera information units **31c** to **34c**.

(86) Also, display processing or the like for superimposing these CGs and characters on the image is performed. Here, the first display unit **50**, the second display unit **51**, and the like function as display units and display image signals and integrated image recognition results.

(87) Furthermore, in the first embodiment, the integrated control unit **41c** is configured to share information on the recognized object among the plurality of camera units. That is, for example, an object recognized by the camera unit **14** is assumed to move in a direction of the angle of view of the camera unit **11**.

(88) In that case, the integrated control unit **41c** transmits prediction information including types of the object and information on a moving direction of the object or information on the priority recognition region to the recognition unit **31b** of the camera unit **11**.

(89) When such prediction information is shared with the recognition units **31b** to **34b** of the

camera units **11** to **14**, an image recognition accuracy of the recognition units **31b** to **34b** of the camera units **11** to **14** can be improved. Further, an advantage of sharing such prediction information is particularly effective when the recognition units **31b** to **34b** of the camera units **11** to **14** are provided separately from the recognition unit **41b** of the integrated processing unit **40**.

(90) Also, the integrated control unit **41c** communicates with the travel control unit (ECU) **60** or the like via a communication unit (not shown) provided inside using a protocol such as CAN, FlexRay, or Ethernet. Thereby, display processing that appropriately changes information to be displayed on the basis of a vehicle control signal received from the travel control unit (ECU) **60** or the like is performed. That is, for example, a range of the image or the like to be displayed on the display unit is changed according to a moving state of the vehicle acquired by the vehicle control signal.

(91) Further, the travel control unit (ECU) **60** is mounted on the vehicle **1** and is a unit in which a computer and a memory for comprehensively performing drive control, direction control, and the like of the vehicle **1** are incorporated. From the travel control unit (ECU) **60**, information on vehicle travel (moving state) such as, for example, a travel speed, a travel direction, a shift lever, a shift gear, a state of turn signals, and a vehicle orientation due to geomagnetic sensors, and the like is input to the integrated processing unit **40** as vehicle control signals.

(92) Conversely, the integrated control unit **41c** transmits information such as a type, a position, a moving direction, and a moving speed of a predetermined object (obstacle or the like) recognized by the recognition unit **41b** to the travel control unit (ECU) **60**. Thereby, the travel control unit (ECU) **60** performs control required for avoiding obstacles such as stopping, driving, or changing a traveling direction of the vehicle. Here, the travel control unit (ECU) **60** functions as a movement control unit that controls movement of the vehicle as the mobile object on the basis of the integrated image recognition result.

(93) The first display unit **50** is installed, for example, in the vicinity of a center in a vehicle width direction on a front upper portion of a driver's seat of the vehicle **1** with the display screen facing the rear of the vehicle, and functions as an electronic rearview mirror. Further, the first display unit **50** may be configured so that it can be used as a mirror by using a semitransparent mirror or the like when it is not used as a display. Also, the first display unit **50** may be configured to include a touch panel and operation buttons to be able to acquire an instruction from the user and output it to the integrated control unit **41c**.

(94) The second display unit **51** is installed, for example, around an operation panel in the vicinity of the center in the vehicle width direction in front of the driver's seat of the vehicle **1**. Further, a navigation system (not shown), an audio system, and the like are mounted in the vehicle **1** as the mobile object.

(95) Then, for example, control signals of various types from the navigation system, the audio system, the travel control unit (ECU) **60**, and the like can be displayed on the second display unit. Also, the second display unit **51** may be configured to include a touch panel and operation buttons to be able to acquire an instruction from the user.

(96) Further, the second display unit **51** may be, for example, a display unit of a tablet terminal, can be displayed by being connected to the integrated processing unit **40** by wire, or can also wirelessly receive and display an image via a communication unit **62**.

(97) Further, as a display panel of the first display unit **50** and the second display unit **51**, a liquid crystal display, an organic EL display, or the like can be used. Further, the number of display units is not limited to two.

(98) Further, a part or all of the functional blocks included in the integrated processing unit **40** or the like may be realized by hardware, or may be realized by causing the CPU **42** to execute a computer program stored in the memory **43**. As the hardware, a dedicated circuit (ASIC), a processor (reconfigurable processor, DSP), or the like can be used.

(99) Further, a part or all of the image processing performed by the image processing units **31a** to

34a may be performed by the image processing unit **41a** of the integrated processing unit **40**. That is, in the first embodiment, for example, the image acquisition unit and the first image recognition unit are housed in the housing of the same camera unit, and the camera unit and the second image recognition unit are housed in separate housings. However, for example, the first image recognition unit may be housed in the housing of the integrated processing unit **40** together with the second image recognition unit.

(100) Further, in the first embodiment, the integrated processing unit **40** is mounted in the vehicle **1** as the mobile object, but a part of the processing of the image processing unit **41a**, the recognition unit **41b**, and the integrated control unit **41c** of the integrated processing unit **40** may be performed by, for example, an external server or the like via a network.

(101) In that case, for example, the imaging units **21** to **24** as the image acquisition units are mounted in the vehicle **1** as the mobile object, but for example, some of the functions of the camera processing units **31** to **34** and the integrated processing unit **40** can be processed by the external server or the like. It is also possible to make the travel control unit (ECU) **60** hold some or all of the functions of the integrated processing unit **40**.

(102) Reference sign **61** denotes a storage unit, which stores the entire image of each of the imaging units **21** to **24** generated by the integrated processing unit **40** and a panoramic synthesized image. Further, a predetermined frame indicating a recognized object, a CG such as characters and warnings, and an image on which the CG is superimposed and displayed on the first display unit **50**, the second display unit **51**, and the like are stored together with a time, GPS information, or the like. The integrated processing unit **40** can reproduce past information stored in the storage unit **61** and display it on the first display unit **50** or the second display unit **51**.

(103) Reference sign **62** denotes a communication unit, which is for communicating with an external server or the like via a network and can transmit information before being stored in the storage unit **61** or past information stored in the storage unit **61** to an external server or the like to store the information in the external server or the like. Also, as described above, an image can be transmitted to an external tablet terminal or the like, and the image can be displayed on the second display unit **51** which is a display unit of the tablet terminal.

(104) Also, it is also possible to acquire traffic congestion information and information of various types from an external server or the like and display them on the first display unit **50** or the second display unit **51** via the integrated processing unit **40**. Reference sign **63** denotes an operation unit for inputting various instructions to the image processing system by an operation of the user. The operation unit includes, for example, a touch panel and operation buttons.

(105) FIG. 4 is a diagram for explaining a correspondence relationship between the angle of view, the high-resolution region, and the low-resolution region of the camera unit **13** on a rear side in the first embodiment. As shown in FIG. 4, an image of the camera unit **13** on the rear side having the imaging angle of view **13a** in which a high-resolution and low-distortion image can be captured is formed in the high-resolution region (low-distortion region) **10a** of a light receiving surface **81** of the imaging element. Also, an image of the imaging angle of view **13b** in which a low-resolution and high-distortion image can be captured is formed in the low-resolution region (high-distortion region) **10b** of the light receiving surface **81** of the imaging element.

(106) FIG. 5A is a diagram showing an example of an image signal obtained from the imaging element when distortion-correction is not performed. As shown in FIG. 5A, distortion is small in the image obtained from the high-resolution region (low-distortion region) **10a**, but the image obtained from the low-resolution region (high-distortion region) **10b** has large distortion.

(107) FIG. 5B is a diagram for explaining an image signal after distortion-correction, and the distortion of the image signal from the low-resolution region (high-distortion region) **10b** is corrected as shown in the figure. However, even if the distortion is corrected, distortion in the peripheral portion of the image may slightly remain.

(108) FIG. 6A is a diagram showing an example in which distortion is corrected for the entire

screen. In this case, a delay is generated even in the low-distortion region (high-resolution region), but it is easier to recognize an object compared to a case before the distortion-correction. Also, the high-distortion region (low-resolution region) is easier to recognize (recognize an image of or visually recognize) the object, but a delay is generated.

(109) FIG. 6B is a diagram showing an example of synthesizing images by performing distortion-correction on an image obtained from the low-resolution region (high-distortion region) **10b** without performing distortion-correction on an image obtained from the high-resolution region (low-distortion region) **10a**. In a case of this example, a delay does not occur in the high-resolution region (low-distortion region), but there is a delay in the low-resolution region (high-distortion region). Since the high-resolution region has low-distortion even without distortion-correction, recognition processing of the object can be performed without a major problem.

(110) Further, the distortion-correction in the first embodiment is correction processing for reducing distortion, and distortion may remain, for example, in a portion such as a peripheral portion including a case in which the distortion does not become zero. Further, in FIG. 6B, images are synthesized by correcting distortion of the image obtained from the low-resolution region (high-distortion region) **10b** to obtain the image **10c** without correcting distortion of the image obtained from the high-resolution region (low-distortion region) **10a**. However, the region without distortion-correction is not limited to the circular high-resolution region (low-distortion region) **10a** as shown in FIG. 6B. The region without distortion-correction may have another shape such as a rectangle, and a size and a position thereof may be changeable.

(111) Further, FIG. 6C is a diagram showing an example of a case in which distortion-correction is not performed for the entire screen. In a case of this example, a delay is not generated in the screen display, but there is a problem that it is difficult to recognize (recognize an image of or visually recognize) the object in a peripheral portion of the imaging region.

(112) In the first embodiment, it is configured such that these three display patterns (a shape, a position, and a size of a region for distortion-correction) are appropriately switched. Further, as described above, a shape, a position, and a size of a region for distortion-correction (or a region without distortion-correction) may be switched.

(113) FIG. 7 is a flowchart for explaining an example of a case in which the pattern of FIG. 6 is manually switched. The flow of FIG. 7 is processed, for example, in units of frame by causing the CPU **42** of the integrated processing unit **40** to execute a computer program in the memory **43** according to a user input from the operation unit **63** or the like. Here, the user may be an operator of the vehicle **1** or an expert who maintains the vehicle.

(114) In FIG. 7, it is determined in step **S71** whether distortion-correction is set to ON by the operation unit **63** or the like. In a case of Yes, the processing proceeds to step **S72**, and it is determined whether a low delay mode is set by the operation unit **63** or the like. If it is determined to be No in step **S72**, the processing proceeds to step **S73**, an image in which the entire screen has been corrected for distortion is selected, and then it is output to the display unit in step **S74**.

(115) In this case, an image in which the entire screen has been corrected for distortion as shown in FIG. 6A is displayed on the first display unit **50** or the second display unit **51**. Further, when the entire screen is corrected for distortion, the image processing unit **41a** performs distortion-correction of the entire screen on the basis of the characteristic information of the optical system acquired from the camera information units **31c** to **34c**.

(116) In a case of Yes in step **S72**, that is, if the low delay mode is set, the processing proceeds to step **S76**. Then, on the basis of the characteristic information of the optical system, the image obtained from the low-resolution region (high-distortion region) **10b** is corrected for distortion while the image obtained from the high-resolution region (low-distortion region) **10a** is not corrected for distortion, and both the images are synthesized.

(117) Then, the synthesized image signal is displayed on the first display unit **50** or the second display unit **51** in step **S77**. The image displayed at this time is the image shown in FIG. 6B.

Further, at this time, it may be configured such that the user can arbitrarily set a position, a size, a shape of the region for distortion-correction (the region without distortion-correction).

(118) In a case of No in step **S71**, that is, if distortion-correction is not set to ON, the processing proceeds to step **S78**, the entire image that has not been corrected for distortion is selected, and then the image signal is displayed on the first display unit **50** or the second display unit **51** in step **S79**. The image displayed at this time is an image in which the entire screen is distorted as shown in FIG. **6C**.

(119) As described above, the flow of FIG. **7** is configured so that the user can select and set which region of the images screen to be corrected for distortion by using the operation unit **63** or the like. Thereby, since the entire image with little distortion, for example, as shown in FIG. **6A** can be displayed in steps **S73** and **S74**, if a high quality image is desired to be displayed, such a selection can be made.

(120) However, in this case, there is a disadvantage in that it takes time to correct the distortion. Also, for example, if it is desired to quickly recognize (recognize an image of or visually recognize) only a central portion, the image as shown in FIG. **6B** can be displayed by steps **S76** and **S77**. In this case, since it does not take time to correct distortion for the central portion, an image of an obstacle or the like can be recognized in a short period of time.

(121) On the other hand, a high-quality image with little distortion can be obtained except for the central portion or the like, and thus an image recognition accuracy improves. Also, if it is desired to immediately recognize an image of the obstacle, an image in which the entire screen is distorted as shown in FIG. **6C** can be quickly displayed by steps **S78** and **S79**. However, a recognition accuracy when the image is recognized is lowered in the peripheral portion.

(122) Further, after the display is performed in steps **S74**, **S77**, and **S79**, the processing proceeds to step **S75** to determine whether or not there has been an end request from the user, and if it is determined as Yes, the flow of FIG. **7** ends, and if it is determined as No, the processing returns to step **S71**.

(123) In the first embodiment, an optimum distortion processing region can be selected by configuring the region for distortion-correction (or the non-distortion-correction region without distortion-correction) to be manually selected by the user using the operation unit **63** or the like. That is, it is possible to appropriately display an image that is easy for the user to recognize with little delay, and it is possible to further bring out performance of the image processing system that uses the optical system such as a different-angle-of-view lens.

(124) Further, in the flow of FIG. **7**, steps **S71** and **S72** using the operation unit **63** or the like function as setting steps (setting units) for setting a distortion-correction region on which distortion-correction is performed for the image signal on the basis of characteristics of the optical image and a non-distortion-correction region without distortion-correction. Also, a state in which the entire screen is set as the distortion-correction region and a state in which the entire screen is set as the non-distortion-correction region can be set by steps **S71** and **S72**.

(125) Also, it is desirable that setting of the distortion-correction region and the non-distortion-correction region without distortion-correction by the user be further facilitated by using a UI such as a menu selection screen in the operation unit **63** or the like.

(126) Further, if the selection or setting is made by the user using the operation unit **63** or the like, the UI may be configured so that the user selects one of, for example, FIGS. **6A** to **6C** from a plurality of menus. Alternatively, for example, FIG. **6B** or the like may be displayed on a screen of an external PC terminal via the communication unit **62**. Then, the UI may be configured such that, for example, a position, a size, and a shape of the region for distortion-correction (the region without distortion-correction) are changed by a touch operation, a click operation, a drag operation, or the like on the screen of the PC terminal.

(127) Here, the circular high-resolution region **10a** as shown in FIG. **6B** has been described as the region without distortion-correction in steps **S76** and **S77** described above, but a position, a size,

and a shape of the region without distortion-correction can be set by the user.

Second Embodiment

(128) Selection of the distortion-correction region has been made manually in the first embodiment, but in a second embodiment, the selection is automatically performed. FIG. 8 is a flowchart for explaining a distortion region selection example in the second embodiment. The flow of FIG. 8 is processed, for example, in units of frame by a CPU 42 of an integrated processing unit 40 executing a computer program in a memory 43 according to a signal indicating a traveling state from a travel control unit (ECU) 60.

(129) In step S81 of FIG. 8, it is determined whether a vehicle speed of a vehicle 1 is equal to or higher than a predetermined speed according to a signal indicating a traveling state from the travel control unit (ECU) 60. If it is determined to be No, the processing proceeds to step S82, and recognition units 31b to 34b recognize the image obtained from a high-resolution region (low-distortion region) 10a, and determine whether or not the object is recognized.

(130) Further, an image recognition region in the recognition units 31b to 34b is, for example, a rectangular region initially set in the high-resolution region 10a in advance. That is, the recognition units 31b to 34b perform image recognition on the basis of a RAW image signal on which distortion-correction has not been performed by a distortion-correction unit.

(131) In a case of No in step S82, that is, if an object is not recognized in the image obtained from the high-resolution region (low-distortion region) 10a, the processing proceeds to step S83, an image that has been corrected for distortion is selected, and the selected image is output to a display unit in step S84. In this case, an image in which the entire screen has been corrected for distortion as shown in FIG. 6A is displayed in a first display unit 50 or a second display unit 51. In a case of Yes in step S82, that is, if an object is recognized in the image obtained from the high-resolution region (low-distortion region) 10a, the processing proceeds to step S86.

(132) Then, an image obtained from a low-resolution region (high-distortion region) 10b is corrected for distortion while the image obtained from the high-resolution region (low-distortion region) 10a is not corrected for distortion, and both the images are synthesized. Then, the synthesized image signal is displayed on the first display unit 50 or the second display unit 51 in step S87. The image displayed at this time is the image as shown in FIG. 6B. As described above, in the flow of FIG. 8, the distortion-correction region and the non-distortion-correction region without distortion-correction are set on the basis of results of the image recognition by the recognition units 31b to 34b as a first image recognition unit.

(133) In a case of Yes in step S81, that is, if a speed of the vehicle 1 is higher than the predetermined speed, the processing proceeds to step S88, the entire image that has not been corrected for distortion is selected, and then the image signal is displayed on the first display unit 50 or the second display unit 51 in step S89. The image displayed at this time is an image in which the entire is distorted as shown in FIG. 6C.

(134) In this way, which region of the imaged screen to be corrected for distortion is automatically selected, and thereby, for example, if the speed of the vehicle 1 is low, the entire image with little distortion as shown in FIG. 6A can be displayed by steps S83 and S84. Therefore, a high-quality image can be displayed. In this case, display is delayed because it takes time to correct the distortion, but delay in the display (image recognition or visual recognition) does not cause a problem because the speed of the vehicle 1 is low.

(135) Also, for example, if an object is recognized in a central portion, an image as shown in FIG. 6B is displayed by steps S86 and S87. Therefore, since it does not take time to correct distortion for the central portion, an obstacle or the like can be recognized in a short period of time, and a high-quality image with little distortion can be obtained except for the central portion or the like.

(136) Also, if the vehicle speed is equal to or higher than the predetermined speed, since an image in which the entire screen is distorted as shown in FIG. 6C can be quickly displayed by steps S88 and S89, an obstacle or the like can be quickly recognized and danger can be avoided. Moreover,

since a peripheral region of the screen is also displayed, a range of image recognition is extended and an obstacle is easily recognized.

(137) Further, after the display is performed in steps **S84**, **S87**, and **S89**, the processing proceeds to step **S85** to determine whether or not there has been an end request from a user, and if it is determined as Yes, the flow of FIG. **8** ends, and if it is determined as No, the processing returns to step **S81**. Further, in the flow of FIG. **8**, steps **S81** and **S82** function as setting steps (setting units) for setting a distortion-correction region on which distortion-correction is performed for the image signal on the basis of characteristics of the optical image and a non-distortion-correction region without distortion-correction.

(138) Also, the distortion-correction region and the non-distortion-correction region without distortion-correction are set according to a moving state of the vehicle **1** as the mobile object by steps **S81** and **S82**.

(139) Further, in the first embodiment and second embodiment, the recognition units **31b** to **34b** perform image recognition on, for example, a rectangular region having a predetermined size on an inner side of the high-resolution region (low-distortion region) **10a**, as described above. On the other hand, image recognition on the entire region corresponding to each display screen as shown in the examples of FIGS. **6A** to **6C** and the like is performed in the recognition unit **41b**.

(140) Further, also in the second embodiment, for example, if it is intended not to perform distortion-correction on a part of the screen as shown in FIG. **6B**, it may be configured such that the user can set a region without distortion-correction (a region for distortion-correction) in advance by using an operation unit **63** or the like. That is, for example, a UI in which the user can select it from a plurality of options of a region without distortion-correction (a region for distortion-correction) may be used.

(141) Alternatively, for example, FIG. **6B** or the like may be displayed in advance on a screen of an external PC terminal or the like via a communication unit **62** so that, for example, a position, a size, or a shape of the region without distortion-correction can be set in advance by a UI of a touch operation, a click operation, a drag operation, or the like.

(142) At that time, in FIG. **6B**, the region without distortion-correction may be, for example, a region having a shape such as a rectangle set on an inner side of the high-resolution region **10a**, and the region for distortion-correction may be a display region other than the rectangular region.

(143) Further, when the user sets a position, a size, and a shape of the region without distortion-correction, it is desirable to display a boundary image between the high-resolution region **10a** and the low-resolution region **10b** on the basis of characteristic information of the optical system acquired from camera information units **31c** to **34c**. That is, since the setting can be made with reference to the boundary image, which portion to be corrected for distortion can be correctly determined.

(144) As described above, in the second embodiment, it is configured such that the region for distortion-correction is automatically selected according to a moving state of the vehicle **1** as the mobile object, and thereby an optimum distortion processing region can be selected more quickly than a case of manual switching by a human. Therefore, it is possible to further bring out performance of the image processing system that uses the optical system such as a different-angle-of-view lens.

(145) Further, in the second embodiment, if the speed of the vehicle **1** is equal to or higher than the predetermined speed, the display is switched to, for example, FIG. **6C**. However, for example, if the speed of the vehicle **1** increases from a state of displaying FIG. **6A** or FIG. **6B**, the region without distortion-correction may be controlled to extend stepwise or continuously from a center of the screen toward a periphery thereof.

(146) Also, a position, a size, and a shape of the region for distortion-correction (region without distortion-correction) may be changed according to a signal (for example, a turn signal operation signal, a steering angle signal, information on a traveling direction of the vehicle based on GPS, or

the like) indicating a traveling state from the travel control unit (ECU) **60**.

(147) For example, in a state of right or left turn, the circular region without distortion-correction in FIG. **6B** may be shifted from the center of the screen to the right or left. Alternatively, in a state of right or left turn, a shape of the region without distortion-correction in FIG. **6B** may be a desired non-circular shape extended to the right or left.

Third Embodiment

(148) Next, FIG. **9** is a flowchart for explaining a distortion region selection example in a third embodiment. The flow of FIG. **9** is processed, for example, in units of frame by a CPU **42** of an integrated processing unit **40** executing a computer program in a memory **43** according to a signal indicating a traveling state from a travel control unit (ECU) **60**.

(149) Also, FIG. **10** is a diagram for explaining an example of a distortion-correction region, a display region, and an image recognition region in the third embodiment FIG. **10A** is a diagram for explaining an example of a display region **82** of a reference angle of view of each of camera units **11**, **12**, and **14**, and FIG. **10B** is a diagram for explaining an example of a display region **83** of a narrow angle of view of each of the camera units **11**, **12**, and **14**.

(150) Also, FIG. **10C** is a diagram for explaining an example of a display region **84** of a wide angle of view of each of the camera units **11**, **12**, and **14**, and FIG. **10D** is a diagram for explaining an example of a display region **85** of a narrow angle of view toward the rear of a camera unit **13**. Also, FIG. **10E** is a diagram for explaining an example of a display region **86** of a wide angle of view toward the rear of the camera unit **13**.

(151) In FIGS. **10A** to **10E**, reference sign **81** denotes a light receiving surface of an imaging element, reference sign **10a** denotes a high-resolution region (low-distortion region) described with reference to FIG. **2**, and reference sign **10b** denotes a low-resolution region (high-distortion region). However, a boundary line between the high-resolution region (low-distortion region) **10a** and the low-resolution region (high-distortion region) **10b** is not displayed in a normally displayed image. However, the boundary line described above may be superimposed and displayed on a screen as necessary.

(152) Further, in this example, a region recognized by a recognition unit **41b** is, for example, the entire of the display region. Also, image recognition regions recognized by recognition units **31b** to **34b** are denoted by, for example, reference signs **821** to **861** in FIGS. **10A** to **10E**.

(153) As shown in FIGS. **10A** to **10E**, a plurality of types of display regions, distortion-correction regions, and image recognition regions can be switched, and these regions are switched on the basis of vehicle control information from the travel control unit (ECU) **60**.

(154) In step **S91** of FIG. **9**, the vehicle control information is acquired from the travel control unit (ECU) **60**. As described above, the vehicle control information includes, for example, information on traveling of a vehicle such as a traveling speed, a traveling direction, a shift lever, a shift gear, and a turn signal.

(155) In step **S92**, it is determined whether or not the vehicle is in a state of moving forward on the basis of the vehicle control information. If it is determined to be moving forward, it is determined in step **S93** whether or not a course is being changed, and if it is determined to be No, it is determined in step **S94** whether or not a forward speed is larger than a predetermined threshold value **V1**.

(156) If it is determined to be No in step **S94**, an image having the reference angle of view is displayed in step **S95**. That is, the image of the display region **82** of the reference angle of view in FIG. **10** is displayed on a first display unit **50** as, for example, an electronic rearview mirror, and then the processing proceeds to step **S101**.

(157) Also, at this time, distortion-correction is not performed for an image in the high-resolution region (low-distortion region) **10a**, and distortion-correction is performed on a region on an inner side of the display region **82** of the reference angle of view and on an outer side of the high-resolution region (low-distortion region) **10a**. Also, the recognition units **31b** to **34b** perform image

recognition on the image recognition region **821**. On the other hand, the recognition unit **41b** performs image recognition for the entire display region **82** of the reference angle of view.

(158) FIG. **11** is a diagram for explaining a display example of the first display unit in the third embodiment. Also, FIG. **11A** is a diagram for explaining a display example of an image of a reference angle of view, FIG. **11B** is a diagram for explaining a display example of an image of a narrow angle of view, and FIG. **11C** is a diagram for explaining a display example of an image of a narrow angle of view. In FIG. **11A**, reference sign **501** denotes a display screen of the first display unit as an electronic rearview mirror.

(159) Also, reference sign **82L** denotes an image of a reference angle of view of the camera unit **14** on a left side, reference sign **82C** denotes an image of a reference angle of view of the camera unit **11** on a front side, and reference sign **82R** denotes an image of a reference angle of view of the camera unit **12** on a right side.

(160) If the image of the display region **82** of the reference angle of view of FIG. **10** is displayed on the first display unit **50**, it is displayed as shown, for example, in FIG. **11A**. That is, the image **82L** of the reference angle of view of the camera unit **14** on the left side, the image **82C** of the reference angle of view of the camera unit **11** on the front side, and the image **82R** of the reference angle of view of the camera unit **12** on the right side are displayed to be aligned in order from the left on the display screen **501** of the first display unit **50**.

(161) FIG. **12A** is a diagram for explaining a display example of the display screen **501** of the first display unit **50** of the third embodiment, and FIG. **12B** is a diagram for explaining a display example of a display screen **511** of a second display unit **51** of the third embodiment.

(162) As shown in FIG. **12A**, if the forward speed is determined to be **V1** or less in step **S94**, the images **82L**, **82C**, and **82R** having the reference angle of view are displayed to be aligned on the display screen **501** of the first display unit **50**.

(163) Further, in FIG. **12A**, reference sign **82B** is an image of the reference angle of view of the camera unit **13** on a rear side and is displayed as a picture-in-picture within the display screen **501** of the first display unit **50**. In the third embodiment, if the forward speed is determined to be **V1** or less in step **S94**, the image **82B** having the reference angle of view is also displayed for the image from the camera unit **13**.

(164) On the other hand, if it is determined in step **S94** that the forward speed is higher than **V1**, the processing proceeds to step **S96**, and the image of the display region **83** having a narrow angle of view shown in FIG. **10B** is displayed on the first display unit **50** as an electronic rearview mirror, and then the processing proceeds to step **S101**. The display region **83** is wider on an upper side and narrower in a left-right direction than the display region **82** of the reference angle of view. At this time, since distortion-correction is not performed for the entire image on an inner side of the display region **82** of the reference angle of view, delay in the image display due to the distortion-correction does not occur.

(165) Therefore, an obstacle can be visually recognized quickly. Also, the recognition units **31b** to **34b** perform image recognition for the image recognition region **831**, and the recognition unit **41b** performs image recognition for the entire display region **83** having a narrow angle of view. The image recognition region **831** is set wider on an upper side and narrower on the left and right sides than the image recognition region **821**. This is because, if the forward speed is high, there is a high likelihood of an obstacle or the like coming from an upper side of the screen, and a likelihood thereof from left and right sides will decrease.

(166) An image **83L** of a narrow angle of view of the camera unit **14** on the left side, an image **83C** of a narrow angle of view of the camera unit **11** on the front side, and an image **83R** of a narrow angle of view of the camera unit **12** on the right side are displayed to be aligned as in FIG. **11B**. As described above, if the forward speed is higher than **V1** (for example, 60 km), a visual sense is narrowed, and thus displaying as in FIG. **11B** makes it easier to quickly visually recognize necessary information.

(167) Also, if it is determined in step S93 of FIG. 9 that the course is being changed, the processing proceeds to step S97 in which the image of the display region 84 having a wide angle of view on left and right sides as shown in FIG. 10 C is displayed. Here, the display region 84 has a width that is larger in the left-right direction than the display region 82 of the reference angle of view and extends downward therefrom.

(168) Also, at this time, distortion-correction is not performed for an image in the high-resolution region (low-distortion region) 10a, and distortion-correction is performed on a region on an inner side of the display region 84 and on an outer side of the high-resolution region (low-distortion region) 10a. Since the forward speed is not so high while the course is being changed, such distortion-correction is performed. Further, the recognition units 31b to 34b perform image recognition on the image recognition region 841. On the other hand, the recognition unit 41b performs image recognition for the entire display region 84 having the wide angle of view.

(169) The image recognition region 841 is set narrower vertically and wider on the left and right sides than the image recognition region 821. This is because the forward speed is low while the course is being changed, and there is a high likelihood that an obstacle is present in the left-right direction rather than the vertical direction of the screen.

(170) Then, an image 84L of a wide angle of view of the camera unit 14 on the left side, an image 84C of a wide angle of view of the camera unit 11 on the front side, and an image 84R of a wide angle of view of the camera unit 12 on the right side are displayed to be aligned on the display screen 501 of the first display unit 50 as shown in FIG. 11C.

(171) Further, in FIGS. 11A to 11C, if there is an overlap region in a part of the images of the three reference angles of view, they may be displayed to be joined together and synthesized as a panoramic image.

(172) Further, for example, if the course is being changed to the left, the three images displayed to be aligned as in FIG. 11C may be displayed to be shifted to the left with respect to a center of the display screen 501. Conversely, if the course is being changed to the right, the three images displayed to be aligned as in FIG. 11C may be displayed to be shifted to the right with respect to the center of the display screen 501. Visibility can be improved by displaying in such a manner.

(173) In this way, in the third embodiment, since the image with a wide angle of view is displayed while the course is being changed, it becomes easier to visually recognize safety of the surroundings. Moreover, since the image in which the angle of view is extended downward is displayed, it becomes easier to visually recognize an obstacle on a road. After step S97, the processing proceeds to step S101.

(174) Further, in FIGS. 11A to 11C, the images from the camera units 11, 12, and 14 are displayed to be aligned or synthesized, but for example, the images from the camera units 12, 13, and 14 may be displayed to be aligned or synthesized.

(175) For example, FIG. 11D is a diagram showing a display example in which the images 82R, 82B, and 82L of the reference region from the camera units 12, 13, and 14 are displayed to be synthesized, and the image 82C of the reference region from the camera unit 11 is displayed on an upper side of the synthesized image. Further, these images are displayed on the display screen 501 of the first display unit 50 in FIG. 11D, but they may be displayed on the display screen 511 of the second display unit 51.

(176) In step S92 of FIG. 9, if it is determined that the vehicle is not moving forward, the processing proceeds to step S98, and it is determined whether or not a reverse speed is higher than a predetermined speed V2 (for example, 10 km). If it is determined to be No, the processing proceeds to step S99, and an image 85B of the display region 85 for backward movement having a narrow angle of view in the vertical direction as shown in FIG. 10D is displayed on the display screen 511 of the second display unit 51, for example, as in FIG. 12B. Thereafter that, the processing proceeds to step S101.

(177) Also, at this time, distortion-correction is not performed for an image in the high-resolution

region (low-distortion region) **10a**, and distortion-correction is performed on a region on an inner side of the display region **85** and on an outer side of the high-resolution region (low-distortion region) **10a**. Since the reverse speed is generally low during backward movement, such distortion-correction is performed. Further, the recognition unit **33b** of the camera unit **13** on the rear side performs image recognition on the image recognition region **851**. On the other hand, the recognition unit **41b** performs image recognition for the entire display region **85**.

(178) The image recognition region **851** is set narrower vertically and narrower also on the left and right sides than the image recognition region **821**. This is because, since the reverse speed is low during backward movement, there is little need for quick image recognition over a wide range.

Therefore, in this case, image recognition by the recognition unit **33b** may not be performed.

(179) FIG. **12B** shows an example of a screen displayed on the display screen **511** of the second display unit **51** when the vehicle **1** moves backward, and for example, a guide **512** for guiding the vehicle **1** in a parking space is displayed to be superimposed. FIG. **13A** is a diagram for explaining a display example of the image **85B** having a narrow angle of view in the vertical direction when the vehicle **1** moves backward in the third embodiment, and the image **85B** having the narrow angle of view in the vertical direction is displayed on the display screen **511** of the second display unit **51**, for example, as in FIG. **13A**.

(180) On the other hand, if it is determined in step **S98** that the reverse speed is higher than **V2**, the processing proceeds to step **S100**, and an image **86B** of the display region **86** for backward movement having a wide angle of view in the vertical direction as shown in FIG. **10E** is displayed on the display screen **511** of the second display unit **51** as shown in FIG. **13B**. Thereafter, the processing proceeds to step **S101**. In step **S101**, it is determined whether or not there has been an end request from the user, and if it is determined as Yes, the flow of FIG. **9** ends, and if it is determined as No, the processing returns to step **S91**.

(181) Also, at this time, distortion-correction is not performed for an image in the high-resolution region (low-distortion region) **10a**, and distortion-correction is performed on a region on an inner side of the display region **86** and on an outer side of the high-resolution region (low-distortion region) **10a**. Since the reverse speed is generally not so high during backward movement, such distortion-correction is performed. Further, the recognition unit **33b** of the camera unit **13** on the rear side performs image recognition on the image recognition region **861**. On the other hand, the recognition unit **41b** performs image recognition for the entire display region **86**.

(182) The image recognition region **861** is set wider vertically and wider also on the left and right sides than the image recognition region **851**. This is because, if the reverse speed is high even during backward movement, it is preferable to perform image recognition quickly for a wider range than the image recognition region **851**. Here, FIG. **13B** is a diagram showing a display example of the image **86B** for backward movement having a wide angle of view in the third embodiment.

(183) As shown in FIG. **10E**, the display region **86** for backward movement having a wide angle of view has an angle of view extended further in the vertical direction than that of the display region **85** for backward movement having a narrow angle of view. This is for displaying a place further to the rear to make it easier to visually recognize an obstacle when the reverse speed is higher than **V2**. Further, a left-right width of the display region **86** for backward movement having a wide angle of view and a left-right width of the display region **85** for backward movement having a narrow angle of view are the same in the third embodiment, but the left-right width of the display region **86** may be smaller than the left-right width of the display region **85**.

(184) Further, for example, in FIGS. **10A**, **10C**, **10D**, and **10E**, distortion-correction is not performed for the high-resolution region **10a** in the image display region, distortion-correction is performed for a region other than the high-resolution region **10a** in the image display region. However, the region on which the distortion-correction is not performed may be, for example, a rectangular region on an inner side of the high-resolution region **10a**. Alternatively, the region on which the distortion-correction is not performed may be, for example, a rectangular region larger

than the high-resolution region **10a** including the high-resolution region **10a**.

(185) As described above, in the third embodiment, since the distortion-correction region and the display region can be appropriately changed according to a moving state of the vehicle **1** as the mobile object, it is possible to further bring out performance of the image processing system that uses the optical system such as a different-angle-of-view lens. Further, since the image recognition region is also changed according to a moving state of the vehicle **1** as the mobile object, the performance of the image processing system can be further brought out.

(186) As described above, in the first to third embodiments, the high-resolution region (low-distortion region) **10a** is configured to have a projection characteristic that is approximate to that of a central projection method ($y=f\times\tan\theta$) or an equidistant projection method ($y=f\times\theta$) of a normal optical system for imaging as described above. Therefore, for example, the image of the high-resolution region (low-distortion region) **10a** displayed on the first display unit **50** or the second display unit **51** has a resolution higher than that of the low-resolution region (high-distortion region) **10b**, and a front side, a lateral side, and a distant place to the rear of the vehicle **1** can be displayed more finely.

(187) Also, since the high-resolution region **10a** has a small optical distortion, the image for the electronic rearview mirror displayed on the first display unit **50** can also be displayed in a state in which the distortion is small, and the driver can visually recognize surroundings of the vehicle with a more natural perspective.

(188) Also, since the high-resolution region **10a** in the embodiment is configured to reduce the optical distortion and image recognition can be performed in a state of a RAW image signal without distortion-correction, a processing load for image recognition can be reduced and image recognition can be performed at high speed. Therefore, an obstacle can be found at an early stage on the basis of the image recognition result, and an action for avoiding the obstacle can be performed in a timely manner. As described above, if the configuration of the third embodiment is used, a great effect can be obtained particularly at the time of high-speed traveling on a highway or the like.

(189) Also, since the distortion-correction region is configured to be changed as appropriate, an obstacle or the like can be quickly ascertained according to a moving state of the mobile object or the surrounding conditions, and performance of the optical system such as a different-angle-of-view lens can be brought out to the maximum.

(190) Also, in the above embodiment, an example of selecting whether to perform distortion-correction or not has been described, but if distortion-correction is not performed, a case in which a distortion-correction factor is lower than a predetermined value **X1** is included. Also, a case of performing the distortion-correction includes a case in which the distortion-correction factor is higher than a predetermined value **X2** (however, **X2** is **X1** or more), and **X1** may be set to, for example, 10% and **X2** may be set to, for example, 90%. Also, it is not limited to selecting two types of distortion-correction factors, and it may be configured such that the distortion-correction factor can be gradually changed. The embodiments of the present invention include such ones.

(191) Further, in the above embodiment, an example in which a plurality of camera units are used has been described, but it is also effective in a system having only one camera unit. Further, a boundary image for showing a boundary between a region for distortion-correction and a region without distortion-correction may be made possible to be displayed, for example, during normal traveling.

(192) Particularly, if a luminance level difference is noticeable due to an image shift at the boundary portion due to a presence or absence of distortion-correction or a difference in exposure timing between the distortion-correction region and the non-distortion-correction region, a boundary image as described above may be displayed to reduce a sense of discomfort. In that case, a line width, a concentration, a color, or the like of the boundary image may also be changed to reduce a sense of discomfort between the distortion-correction region and the non-distortion-

correction region.

(193) Also, it may be configured such that the distortion-correction factor in the distortion-correction region is adjusted according to a size and shape of the boundary image to smoothly connect images of a stationary object in the distortion-correction region and the non-distortion-correction region.

(194) Also, the boundary image may be displayed if the user sets a region without distortion-correction in advance other than during normal traveling, or in adjusting a posture of each camera unit. Thereby, the boundary between the distortion-correction region and the non-distortion-correction region becomes clear, and adjustment work can be made more efficient.

(195) Further, in the above-described embodiment, an example in which an image processing system is mounted on the mobile object such as a vehicle has been described. However, the mobile object of these embodiments is not limited to a vehicle such as an automobile, and may be any moving device such as a train, a ship, an airplane, a robot, or a drone.

(196) Also, the image processing system of the embodiment may or may not be mounted on those moving bodies. Also, configurations of these embodiments can be applied to, for example, a case in which the mobile object is controlled remotely.

Fourth Embodiment

(197) At least one of the various functions, processes, and methods described in the first to third embodiments described above can be realized using a program. Hereinafter, in a fourth embodiment, a program for realizing at least one of the various functions, processes, and methods described in the first embodiment described above is referred to as “Program X.”

(198) Further, in the fourth embodiment, a computer for executing the program X is called “computer Y.” A personal computer, a microprocessor, a central processing unit (CPU), and the like are examples of the computer Y. A computer such as the image processing system in the above-described embodiment is also an example of the computer Y.

(199) At least one of the various functions, processes, and methods described in the first to third embodiments described above can be realized by the computer Y executing the program X. In this case, the program X is supplied to the computer Y via a computer-readable storage medium.

(200) The computer-readable storage medium in the fourth embodiment includes at least one of a hard disk device, a magnetic storage device, an optical storage device, a magneto-optical storage device, a memory card, a ROM, a RAM, and the like. Further, the computer-readable storage medium in the fourth embodiment is of a non-transitory storage medium.

(201) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation to encompass all such modifications and equivalent structures and functions.

(202) This application claims the benefit of Japanese Patent Application No. 2021-155137 filed on Sep. 24, 2021, which is hereby incorporated by reference herein in its entirety.

Claims

1. An image processing system comprising: at least one processor or circuit configured to function as a plurality of units comprising: (1) an image acquisition unit configured to acquire an image signal generated by an imaging device that captures an optical image having a low-distortion region and a high-distortion region; (2) a setting unit configured to set a distortion-correction region on which distortion-correction is performed for the image signal and a non-distortion-correction region on which distortion-correction is not performed for the image signal on the basis of characteristics of the optical image; and (3) a display signal generation unit configured to perform distortion-correction for the image signal of the distortion-correction region on the basis of the characteristics of the optical image, and to generate a synthesized image by synthesizing the image

signal on which distortion-correction has been performed and the image signal of the non-distortion-correction region, wherein the setting unit is able to set (1) a state in which an entire screen is set as the distortion-correction region and (2) a state in which an entire screen is set as the non-distortion-correction region.

2. The image processing system according to claim 1, wherein the setting unit comprises a UI for setting the distortion-correction region or the non-distortion-correction region by a user.

3. The image processing system according to claim 1, further comprising at least one processor or circuit configured to function as a characteristics information holding unit configured to hold characteristic information on the characteristics of the optical image, and wherein the display signal generation unit performs the distortion-correction on the basis of the characteristic information of the optical image held by the characteristics information holding unit.

4. The image processing system according to claim 1, wherein the imaging device is installed in a mobile object, and wherein the setting unit sets the distortion-correction region and the non-distortion-correction region according to a moving state of the mobile object.

5. The image processing system according to claim 1, further comprising at least one processor or circuit configured to function as a first image recognition unit configured to perform image recognition on the basis of the image signal, and wherein the setting unit sets the distortion-correction region and the non-distortion-correction region on the basis of a result of image recognition by the first image recognition unit.

6. The image processing system according to claim 5, wherein the first image recognition unit performs image recognition on the basis of an image on which distortion-correction has not been performed.

7. The image processing system according to claim 6, further comprising at least one processor or circuit configured to function as a second image recognition unit configured to perform image recognition on the basis of the synthesized image.

8. The image processing system according to claim 5, wherein the imaging device is installed on a mobile object, and wherein the first image recognition unit changes an image recognition region according to a moving state of the mobile object.

9. The image processing system according to claim 1, wherein the low-distortion region and the high-distortion region respectively correspond to a high-resolution region and a low-resolution region of the optical image.

10. An image processing system comprising: at least one processor or circuit configured to function as a plurality of units comprising: (1) an image acquisition unit configured to acquire an image signal generated by an imaging device that captures an optical image having a low-distortion region and a high-distortion region; (2) a setting unit configured to set a distortion-correction region on which distortion-correction is performed for the image signal and a non-distortion-correction region on which distortion-correction is not performed for the image signal on the basis of characteristics of the optical image; and (3) a display signal generation unit configured to perform distortion-correction for the image signal of the distortion-correction region on the basis of the characteristics of the optical image, and to generate a synthesized image by synthesizing the image signal on which distortion-correction has been performed and the image signal of the non-distortion-correction region, wherein the image acquisition unit includes (1) an optical system that forms the optical image, and (2) an imaging element that captures the optical image formed by the optical system, and wherein, when a focal length of the optical system is f , a half angle of view is θ , an image height on an imaging plane is y , and a projection characteristic representing a relationship between the image height y and the half angle of view θ is $y(\theta)$, $y(\theta)$ in the low-distortion region is larger than $f \times \theta$, and is different from the projection characteristic in the high-distortion region.

11. The image processing system according to claim 10, wherein the low-distortion region has a projection characteristic which is approximate to that of a central projection method ($y=f \times \tan \theta$) or an equidistant projection method ($y=f \times \theta$).

12. The image processing system according to claim 10, wherein, when $\theta_{\max}$ is a maximum half angle of view of the optical system and A is a predetermined constant, the image processing system is configured to satisfy $1 < \frac{f \times \sin \theta_{\max}}{y(\theta_{\max})} \leq A$.

13. An image processing system comprising: at least one processor or circuit configured to function as a plurality of units comprising: (1) an image acquisition unit configured to acquire an image signal generated by an imaging device that captures an optical image having a low-distortion region and a high-distortion region; (2) a setting unit configured to set a distortion-correction region on which distortion-correction is performed for the image signal and a non-distortion-correction region on which distortion-correction is not performed for the image signal on the basis of characteristics of the optical image; and (3) a display signal generation unit configured to perform distortion-correction for the image signal of the distortion-correction region on the basis of the characteristics of the optical image, and to generate a synthesized image by synthesizing the image signal on which distortion-correction has been performed and the image signal of the non-distortion-correction region, wherein the image acquisition unit includes a plurality of the imaging devices disposed at different positions, wherein the display signal generation unit synthesizes the image signals acquired from the plurality of the imaging devices to generate a synthesized image, wherein imaging ranges of at least two imaging devices of the plurality of imaging devices are disposed to overlap each other, and wherein the imaging ranges of the low-distortion regions of at least two imaging devices of the plurality of imaging devices are disposed to overlap each other.
